CITY UNIVERSITY OF HONG KONG

Personal Statement

MSc in Computer Science (P53)

Applicant: LAI Yongyi

While developing an OpenHarmony smart-gardening system, I used YOLOv5 on an RK3588 board to control a pump. I learned reliability depends on timestamped observations, rule-based safety overrides and a Hi3861 state machine as well as model accuracy. I seek CityUHK's MSc in Computer Science to deepen my software engineering, data engineering and AI foundations for robotics.

At Shenzhen Technology University, I am a final-year Computer Science and Technology student ranked 6th among 156 students in my major. At the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), I developed balance control for a three-omni-wheel ballbot, using IMU/wheel feedback with cascaded PID. I used LQR to analyse stability and estimated arm-induced centre-of-mass shifts with URDF/Pinocchio for gravity compensation. In my Kaihong M-Robots internship, I contribute to an OpenHarmony platform for heterogeneous robots: capability abstraction, ROS/ROS2 command mapping, state synchronisation, and M-DDS discovery and task dispatch. Mobile-navigation and dual-arm projects taught me to connect algorithms with hardware, exposing gaps in machine learning and data engineering.

CityUHK's core courses in Software Engineering, Data Engineering and Computer Networks and Internets match challenges I have faced building robot platforms. Machine Learning: Principles and Practice in the Artificial Intelligence Stream would strengthen my theoretical grounding. Through the Project course, I hope to apply this knowledge to robotics perception, navigation or platform software, developing methods transferable across robot types. I expect rigorous coursework and project work to help me approach these systems more systematically.

After graduation, I plan to develop robotics algorithms in the Greater Bay Area, focusing on mobile-robot perception, navigation and motion control. I aim to connect algorithms with dependable platforms and safe robot execution.